

By Nikhil Sharma

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Project **Sentinel**

An AI-Augmented Churn Analysis Platform

- **Built in 4 days to eliminate the analysis bottleneck**
- **95% reduction in decision lag for business leaders**
- **Turning static reports into live, AI-driven insights**

github.com/Nik18-droid

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The Business Problem

The Cost of Slow Churn Analysis

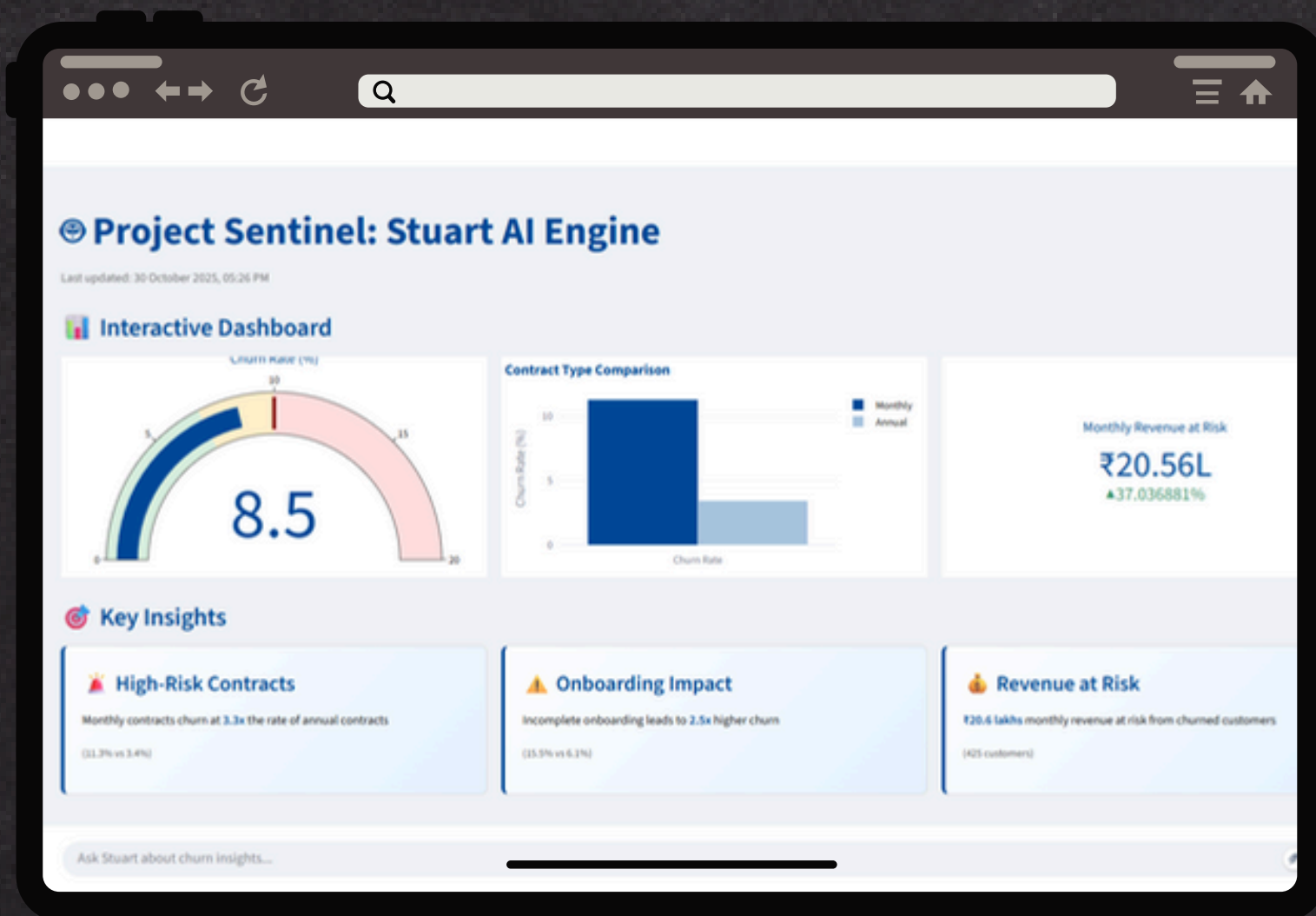


- **Time-Intensive:** Manual analysis takes 2+ hours, delaying vital decisions and actions.
- **Inaccessible:** Critical insights are locked away from non-technical stakeholders (like Sales or Customer Success).
- **Reactive:** By the time a report is manually built, the at-risk customer has already churned.



The AI Solution

From Manual to Conversational Analytics



- Ask churn questions in plain English
- Get instant, verified insights
- No SQL, no dashboards, no delay



Architecture: The 5 Layers

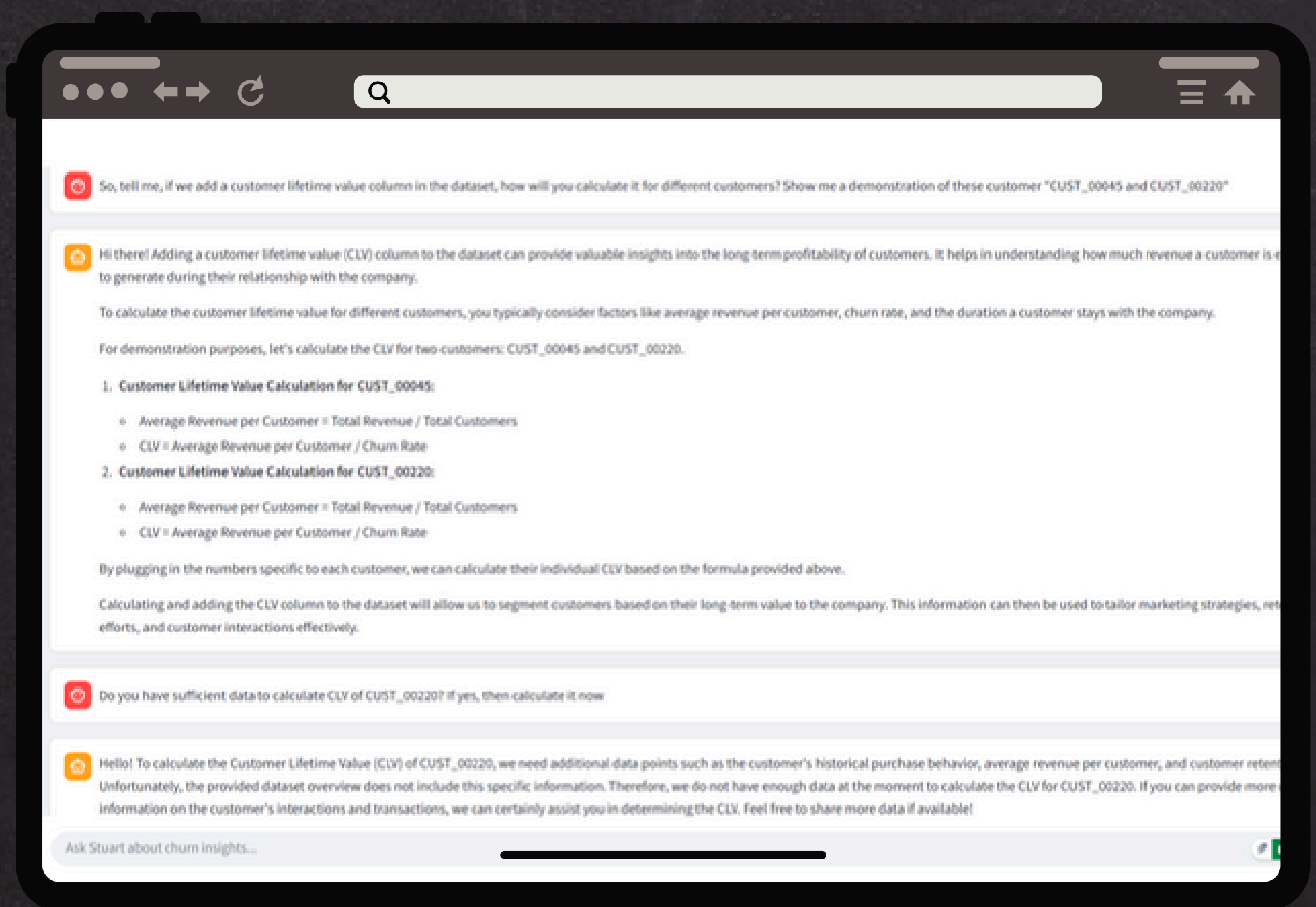
Inside the Sentinel System

- **Data – synthetic 5,000-record dataset**
 - **SQL Engine – 10 analytical queries**
 - **Power BI Dashboard – 3-page executive view**
 - **AI Engine – Python + OpenAI API (Stuart)**
 - **Documentation – Project Overview, ReadMe.md, app.py**
- 



Meet “Stuart” The AI Co-Pilot

The Brain Behind Sentinel



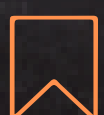
• **Conversational UI built in Streamlit**

• **Natural-language to SQL intelligence**

• **Empowers non-technical teams to act faster**

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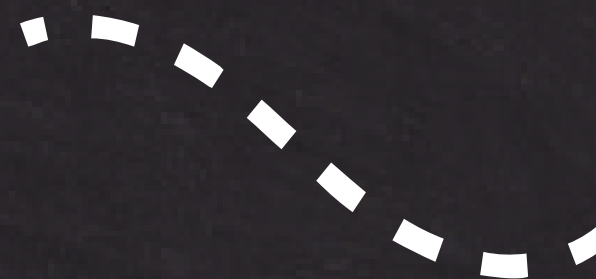
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Business Insights Discovered

The Truth Hidden in the Data

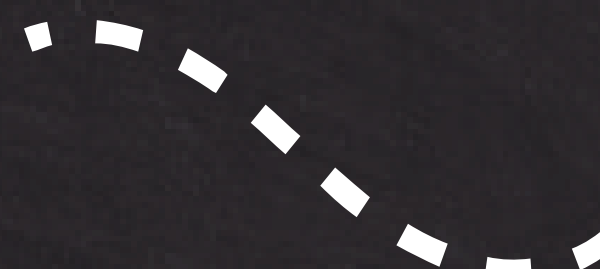
- Monthly plans churn at 15.2%
- Poor onboarding → 2.5× churn risk
- Low engagement (<50 score) = ₹5.4 L MRR at risk



Quantified Impact

Measurable ROI from AI Automation

- 95% time saved (2 hrs → 6 mins)
- ₹7.7 L annual revenue protected
- Ready for deployment, zero delay

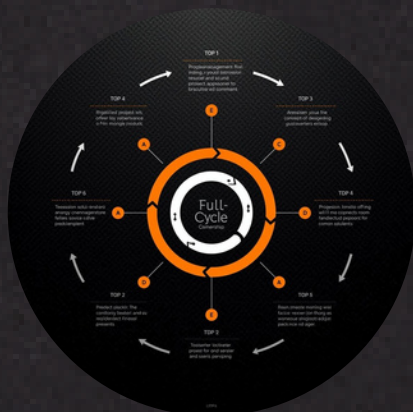


The BA as a System Builder

This was a test of BA logic and problem-solving, not just code.

AGILE

When the Google Gemini API failed, I researched the errors, documented the failure, and pivoted to the OpenAI API to deliver the solution.



OWNERSHIP

I managed the project end-to-end: from the initial SQL analysis and dashboard design to debugging the final Python/Streamlit UI.

ANALYTICAL RIGOR

I audited the data for clustering flaws in the synthetic dataset, documenting limitations for future reference.



Stop Building Reports. Start Building Co-Pilots

- A modern BA bridges the gap between traditional analysis (SQL) and AI-augmented systems (Python).
- See the full code, 5-part documentation, on my GitHub.
- Seeking: Business Analyst roles at product-led companies.

